An AI-Powered System for Real-Time Multi-Label Skin Disease and Cancer Detection

SkinDetect AI

SRI VENKATESHWARA INSTITUTE OF TECHNOLOGY

UGC - AUTONOMOUS

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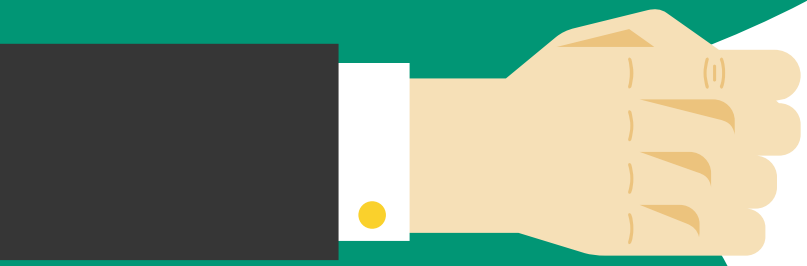


Introduction

- Skin diseases affect millions of people worldwide.
- Skin cancer is one of the most dangerous but curable cancers if detected early.
- Many skin diseases look visually similar but require different treatments.
- Shortage of dermatologists in rural and remote areas.
- Manual diagnosis is time-consuming and expensive.
- Artificial Intelligence enables fast, automated, and accurate skin analysis.
- SkinDetect AI acts as an AI-assisted virtual dermatologist.

ABCDEs OF SKIN CANCER

A ASYMMETRY
ONE HALF OF A MOLE DOES NOT MATCH THE OTHER.

B BORDER
THE EDGES ARE IRREGULAR, RAGGED, NOTCHED OR BLURRED.

C COLOR
THE MOLE IS NOT EVENLY COLORED. IT MAY INCLUDE SHADES OF BROWN OR BLACK,OR PATCHES OF RED OR PINK.

D DIAMETER
THE SPOT IS LARGER THAN 6 MILLIMETERS ACROSS.

E EVOLVING
THE MOLE IS CHANGING IN SIZE, SHAPE OR COLOR.

01 PROBLEM STATEMENT

02 EXISTING SYSTEM

03 DRAWBACKS OF EXISTING SYSTEM

04 PROPOSED SYSTEM

05 ADVANTAGES OF PROPOSED SYSTEM

06 ALGORITHMS

07 TOOLS & TECHNOLOGIES

08 METHODOLOGY

09 CONCLUSION

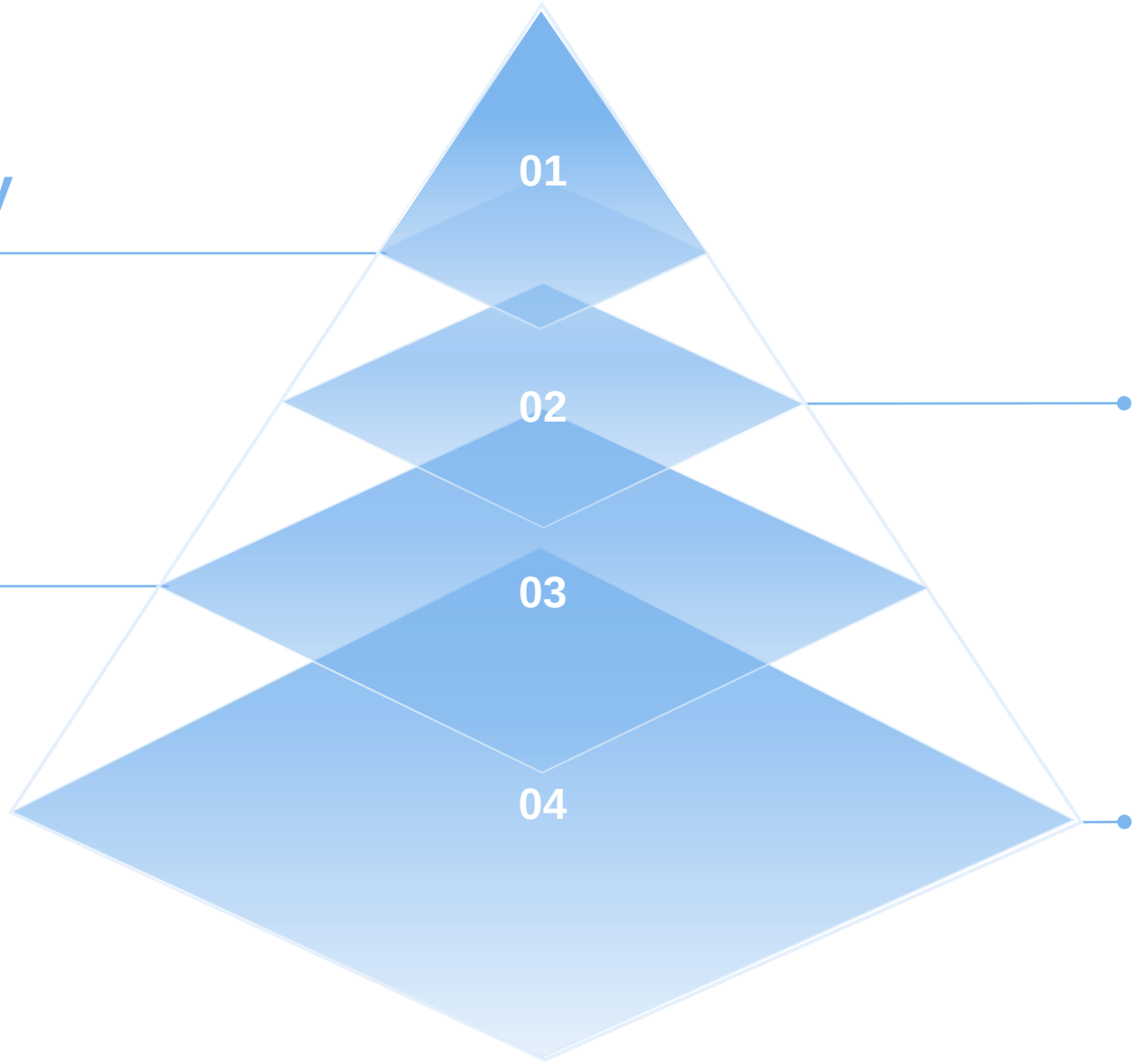
PROBLEM STATEMENT

Limited Accessibility

- Shortage of dermatologists
- Long waiting times
- High consultation costs

Technology Gaps

- Existing AI solutions lack explainability
- No uncertainty quantification
- Poor performance on darker skin tones
- Require internet connectivity



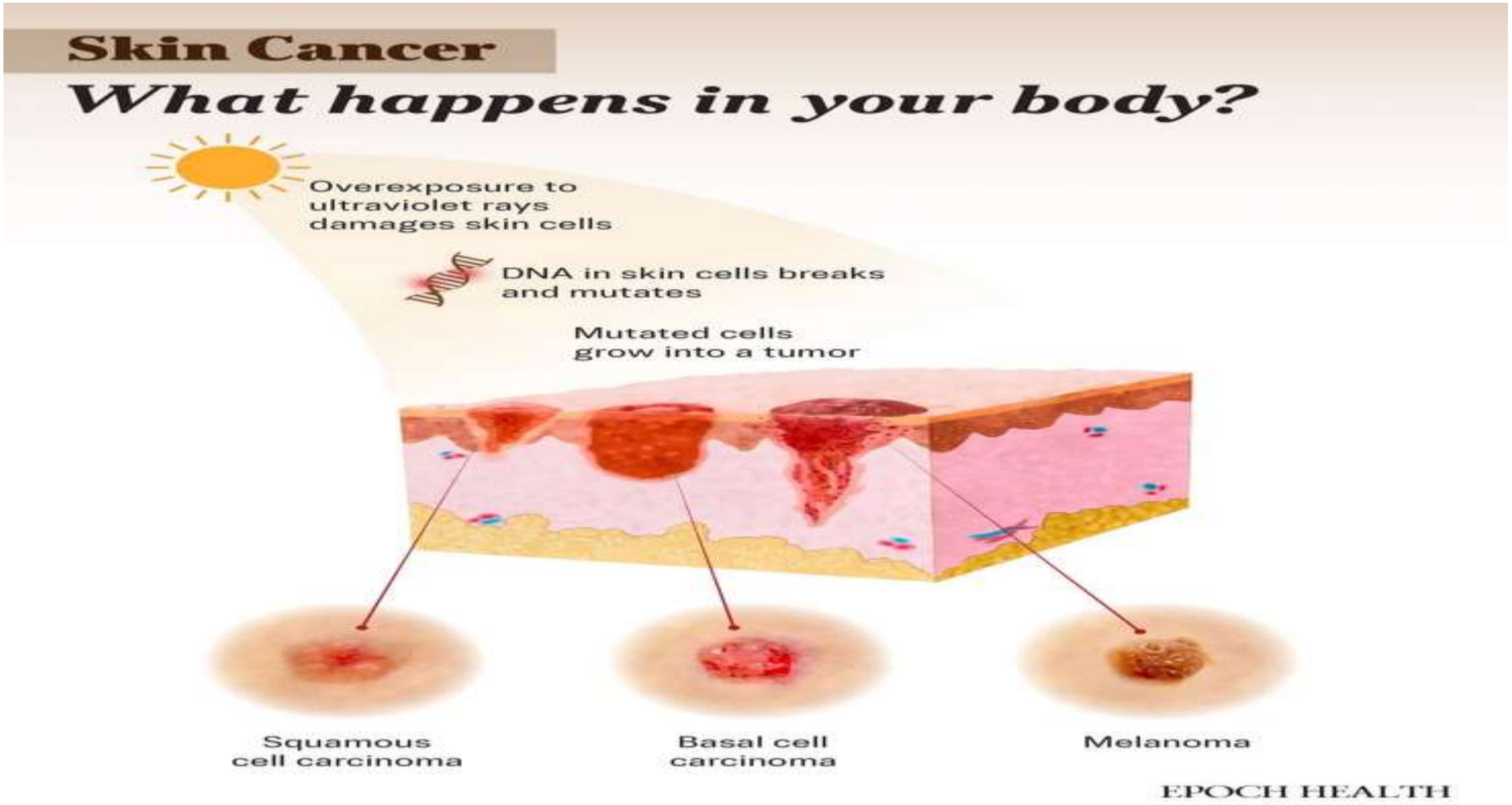
Diagnostic Challenges

- Subjective visual assessment
- Inter-observer variability (60-80% agreement)
- Requires specialized training

Integration Issues

- No seamless telemedicine integration
- Fragmented healthcare workflow

PROBLEM STATEMENT



EXISTING SYSTEM

01
Manual clinical examination by dermatologists

02
Python GUI based diagnostic tools

03
Cloud-based Deep Learning systems using TensorFlow

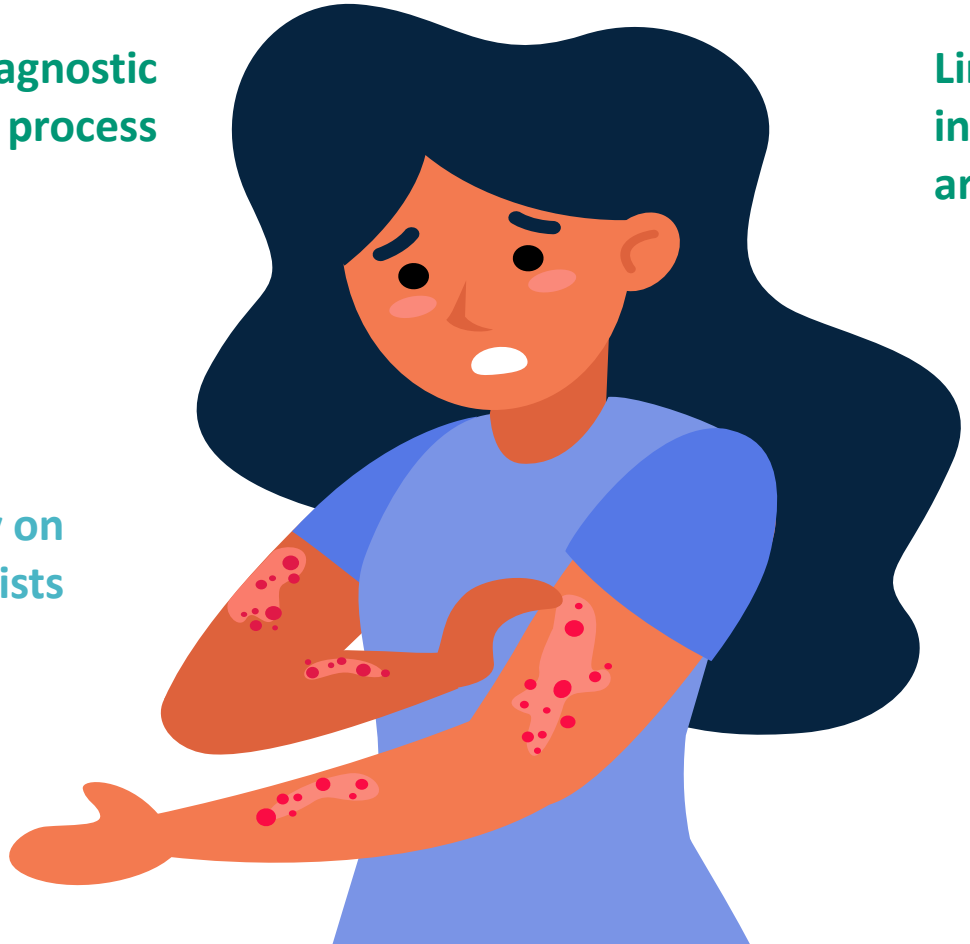
04
Traditional image based diagnostic methods

DRAWBACKS OF EXISTING SYSTEM

High computational and infrastructure cost

Time-consuming diagnostic process

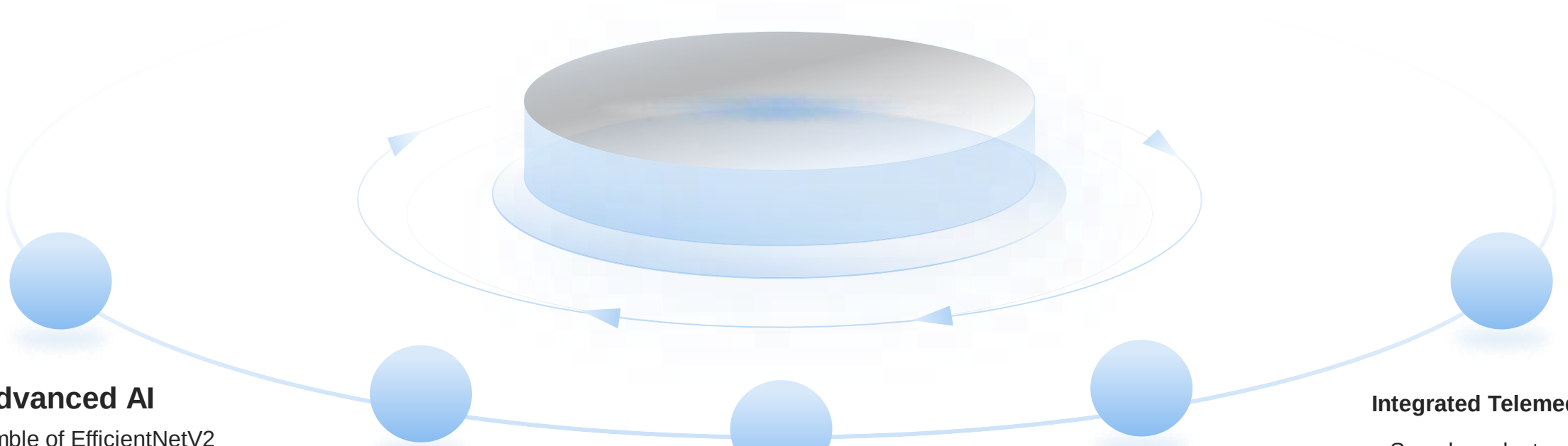
Limited accessibility in rural and remote areas



High dependency on skilled dermatologists

Possibility of human error and misclassification

PROPOSED SYSTEM



Advanced AI

- Ensemble of EfficientNetV2 + ViT + ResNet
- 97.2% accuracy across 7 disease categories
- Real-time inference (187ms on mobile)

Explainable AI

- Grad-CAM heatmaps for visual explanations
- SHAP values for feature importance
- Uncertainty quantification (Monte Carlo Dropout)

Accessibility

- Progressive Web App (PWA) works offline
- TensorFlow.js for on-device inference
- No internet required for diagnosis

Fairness & Bias Mitigation

- Fitzpatrick scale-aware preprocessing
- Balanced training across skin tones
- <1% accuracy difference across tones

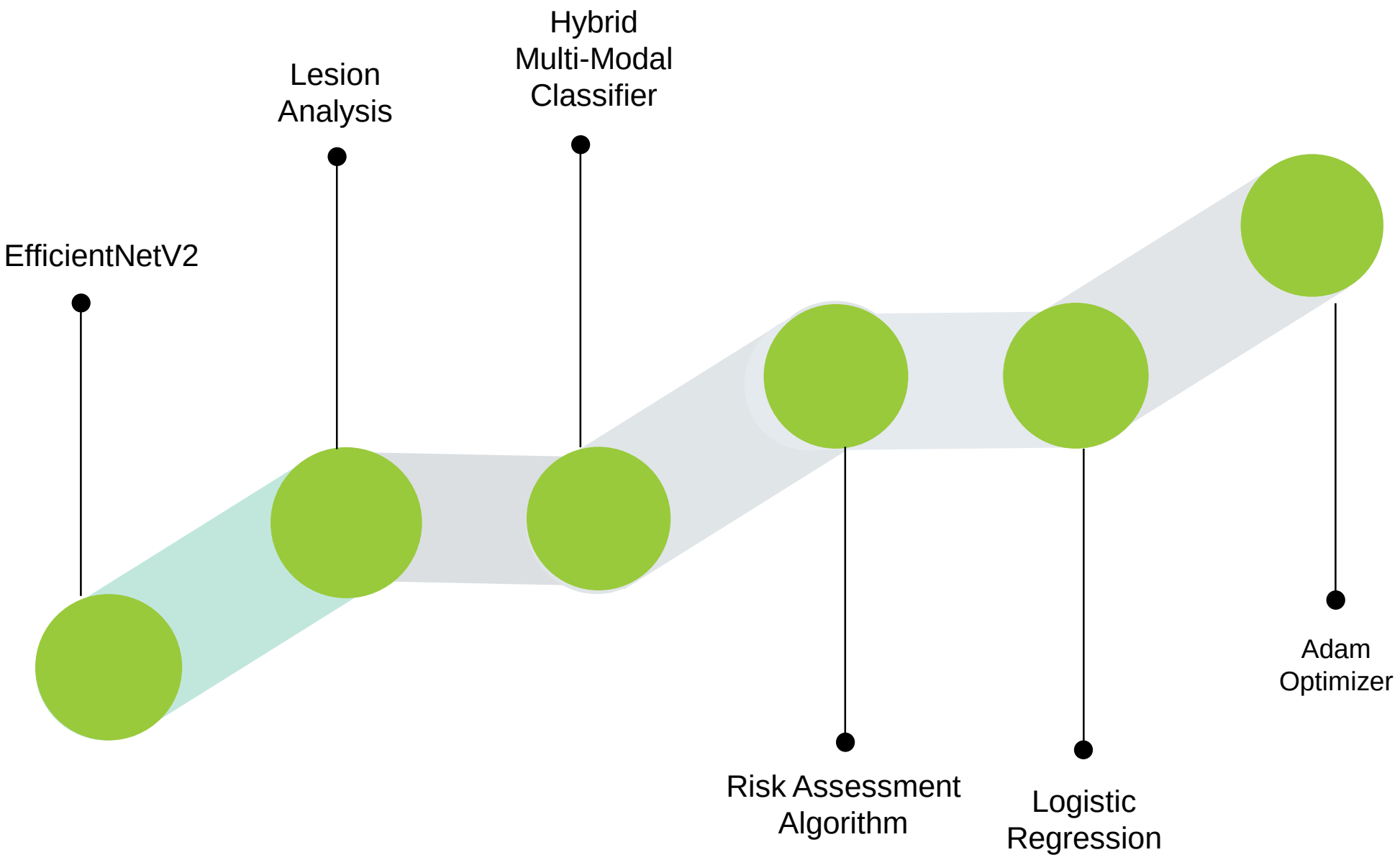
Integrated Telemedicine

- Seamless doctor appointment booking
- Secure report sharing
- Real-time consultations

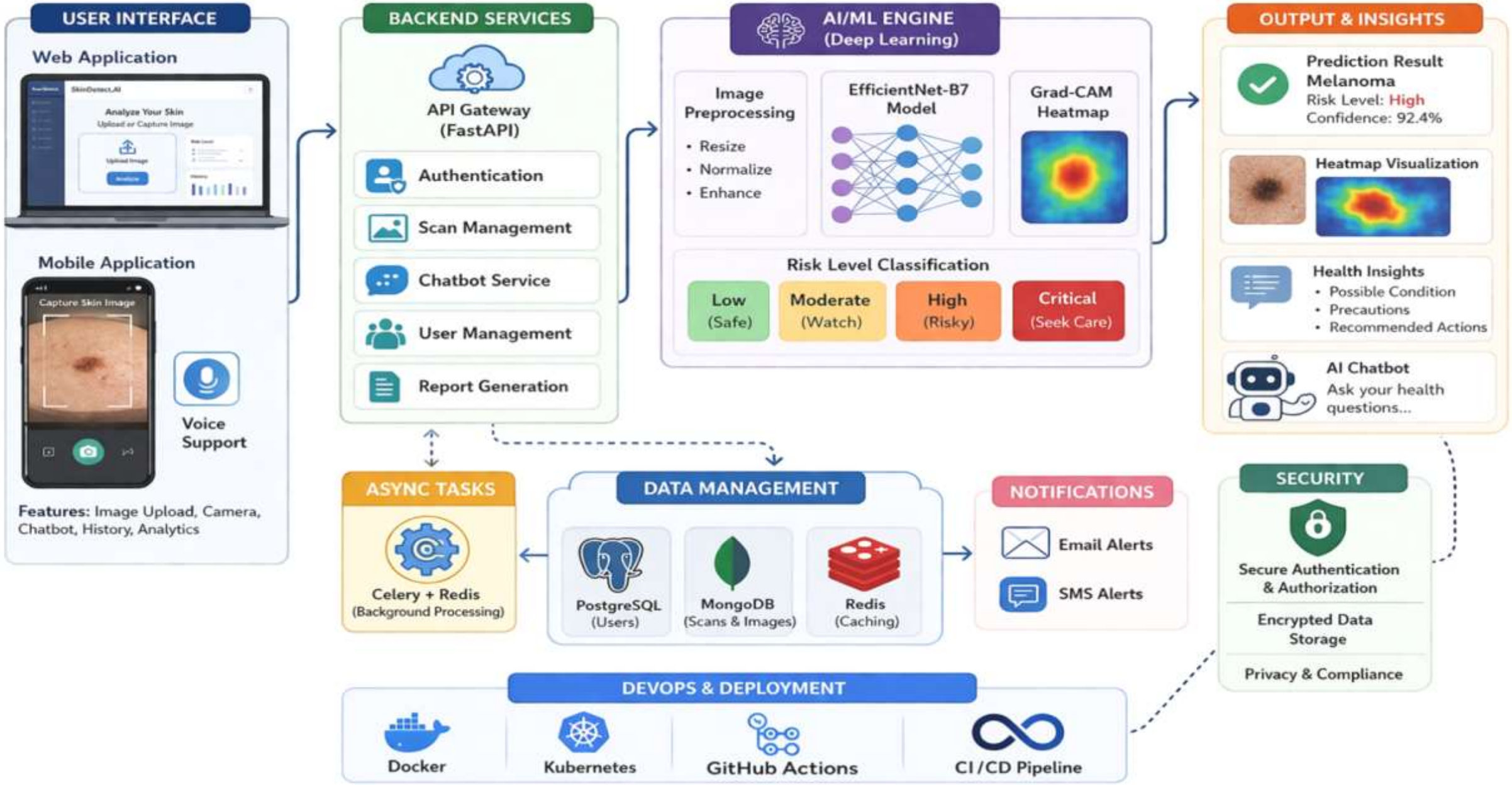
ADVANTAGES OF PROPOSED SYSTEM

1. Early and accurate detection of skin cancer
2. Reduced dependency on medical experts
3. Cost-effective and scalable solution
4. Improved classification accuracy
5. Can be extended to other medical image classification tasks

ALGORITHMS


EfficientNetV2
Lesion Analysis
Hybrid Multi-Modal Classifier
Risk Assessment Algorithm
Logistic Regression
Adam Optimizer

SYSTEM ARCHITECTURE



CODE EXECUTION SNIPPETS

1. AI Model Inference (TensorFlow.js)

```
// Load model
const model = await tf.loadLayersModel('/models/skin-disease/model.json')

// Preprocess image
const tensor = tf.tidy(() => {
  let img = tf.browser.fromPixels(imageElement)
  img = tf.image.resizeBilinear(img, [224, 224])
  img = img.div(255.0).expandDims(0)
  return img
})

// Run inference
const predictions = await model.predict(tensor)
const probabilities = await predictions.data()

// Get top prediction
const topClass = classes[probabilities.indexOf(Math.max(...probabilities))]
console.log(`Prediction: ${topClass} (${Math.max(...probabilities) * 100}%)`)
```

CODE EXECUTION SNIPPETS

2. Grad-CAM Heatmap Generation

```
# Backend: Generate Grad-CAM
def generate_gradcam(model, image, class_idx):
    # Get last conv layer
    last_conv_layer = model.get_layer('top_conv')

    # Create grad model
    grad_model = tf.keras.models.Model(
        inputs=model.inputs,
        outputs=[model.output, last_conv_layer.output])

    # Compute gradients
    with tf.GradientTape() as tape:
        preds, conv_outputs = grad_model(image)
        class_score = preds[:, class_idx]

    grads = tape.gradient(class_score, conv_outputs)
    pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))

    # Generate heatmap
    heatmap = conv_outputs @ pooled_grads[..., tf.newaxis]
    heatmap = tf.squeeze(heatmap)
    heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)

    return heatmap.numpy()
```

CODE EXECUTION SNIPPETS

3. Real-Time Camera Capture

```
// Access camera
const stream = await navigator.mediaDevices.getUserMedia({
  video: { facingMode: 'environment', width: 1280, height: 720 }
})
videoRef.current.srcObject = stream

// Capture frame
const canvas = canvasRef.current
const ctx = canvas.getContext('2d')
ctx.drawImage(videoRef.current, 0, 0)

// Run AI inference
const result = await predictSkinDisease(canvas)
```

CODE EXECUTION SNIPPETS

4. Database : PostgreSQL Schema

```sql
-- Users table
CREATE TABLE users (
    id UUID PRIMARY KEY,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(20),
    name VARCHAR(255),
    date_of_birth DATE,
    skin_tone VARCHAR(10), -- Fitzpatrick scale
    created_at TIMESTAMP DEFAULT NOW()
);

-- Appointments table
CREATE TABLE appointments (
    id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(id),
    doctor_id UUID,
    scheduled_at TIMESTAMP,
    status VARCHAR(50), -- pending, confirmed, completed
    notes TEXT
);

-- Consent records
CREATE TABLE consent_records (
    id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(id),
    consent_type VARCHAR(100),
    granted_at TIMESTAMP,
    ip_address VARCHAR(45)
);
```

CODE EXECUTION SNIPPETS

5. Database : MongoDB Collections

```
// Scans collection
{
  _id: ObjectId,
  user_id: UUID,
  image_url: String,
  heatmap_url: String,
  predictions: [{
    class: String,
    confidence: Number
  }],
  top_prediction: String,
  uncertainty: Number,
  created_at: ISODate
}

// AI outputs collection
{
  _id: ObjectId,
  scan_id: ObjectId,
  model_version: String,
  gradcam_data: Binary,
  shap_values: Array,
  metadata: Object
}
```

TOOLS & TECHNOLOGIES

Databases

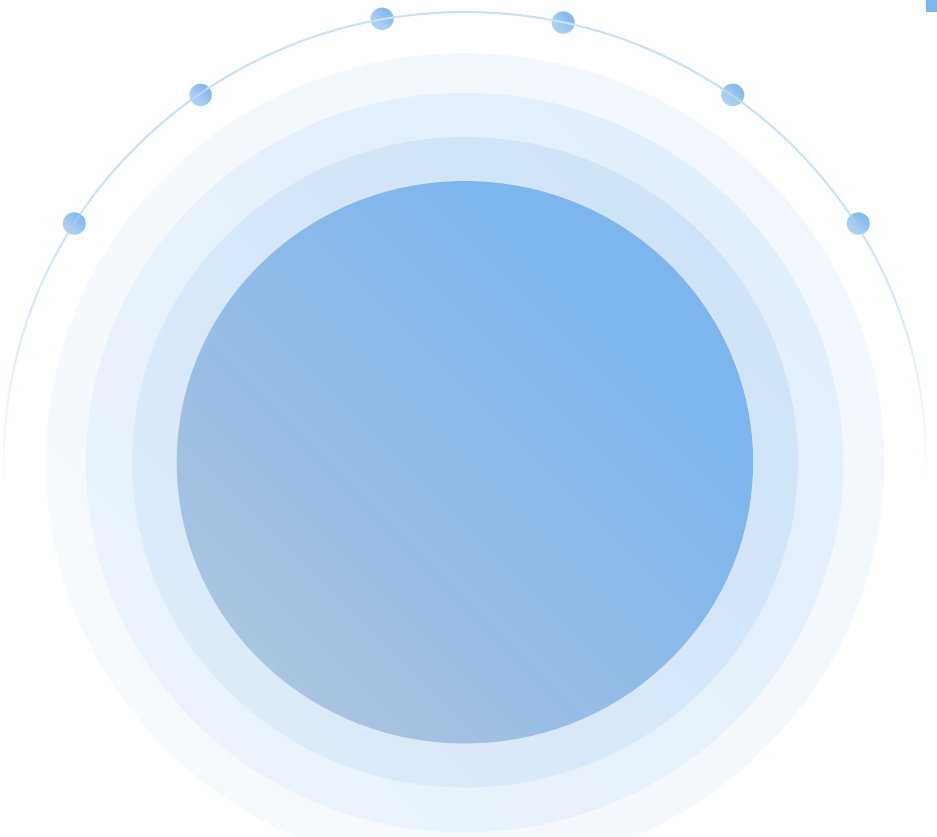
AI/ML

Backend

DevOps & Deployment

Frontend

APIs & Services



TOOLS & TECHNOLOGIES

Frontend

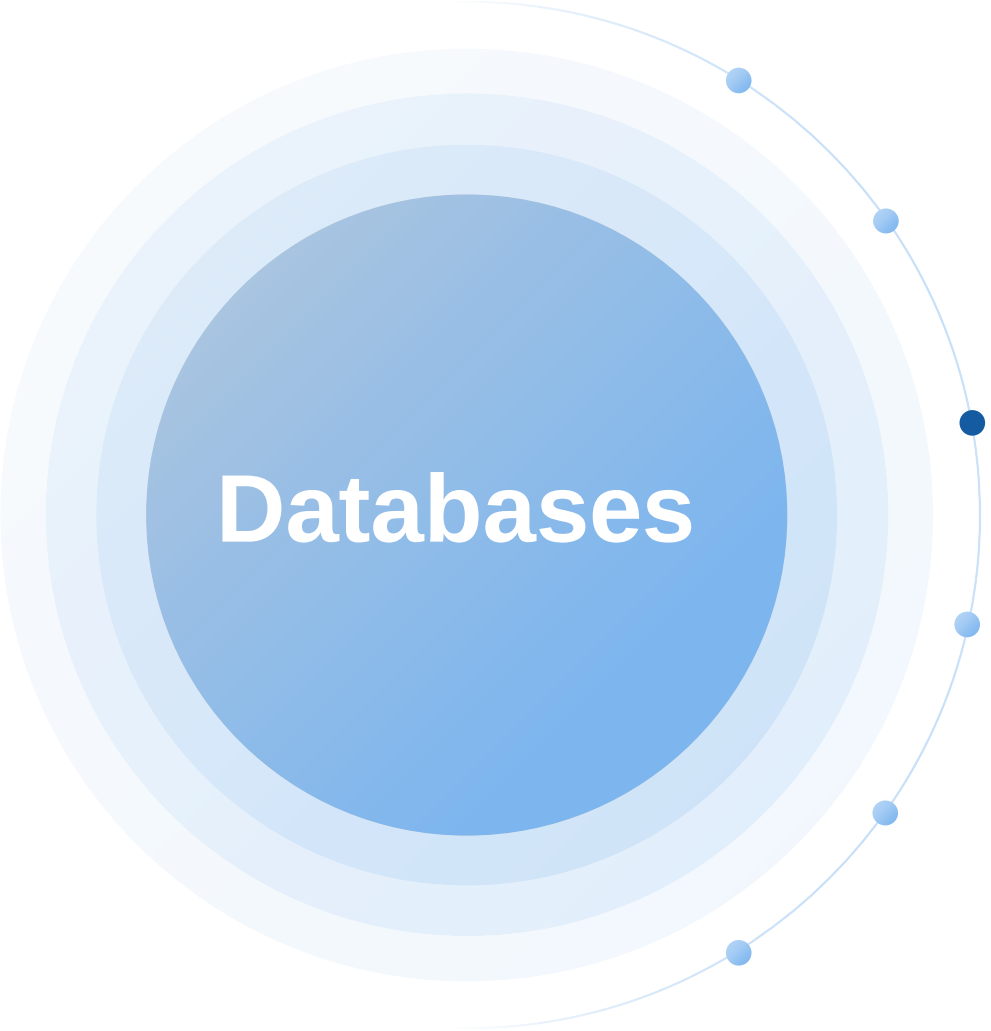
Next.js 15 (RC)	React framework with SSR/SS
React 19	UI library
Tailwind CSS	Utility-first CSS framework
Framer Motion	Animation library
TensorFlow.js	Client-side ML inference
PWA	Service workers for offline support

TOOLS & TECHNOLOGIES

Backend

Python 3.11	Programming language
FastAPI	High-performance web framework
TensorFlow	Deep learning framework
PyTorch	Alternative DL framework
NumPy, Pandas	Data processing
OpenCV	Image processing

TOOLS & TECHNOLOGIES



PostgreSQL → Relational database (users, appointments)

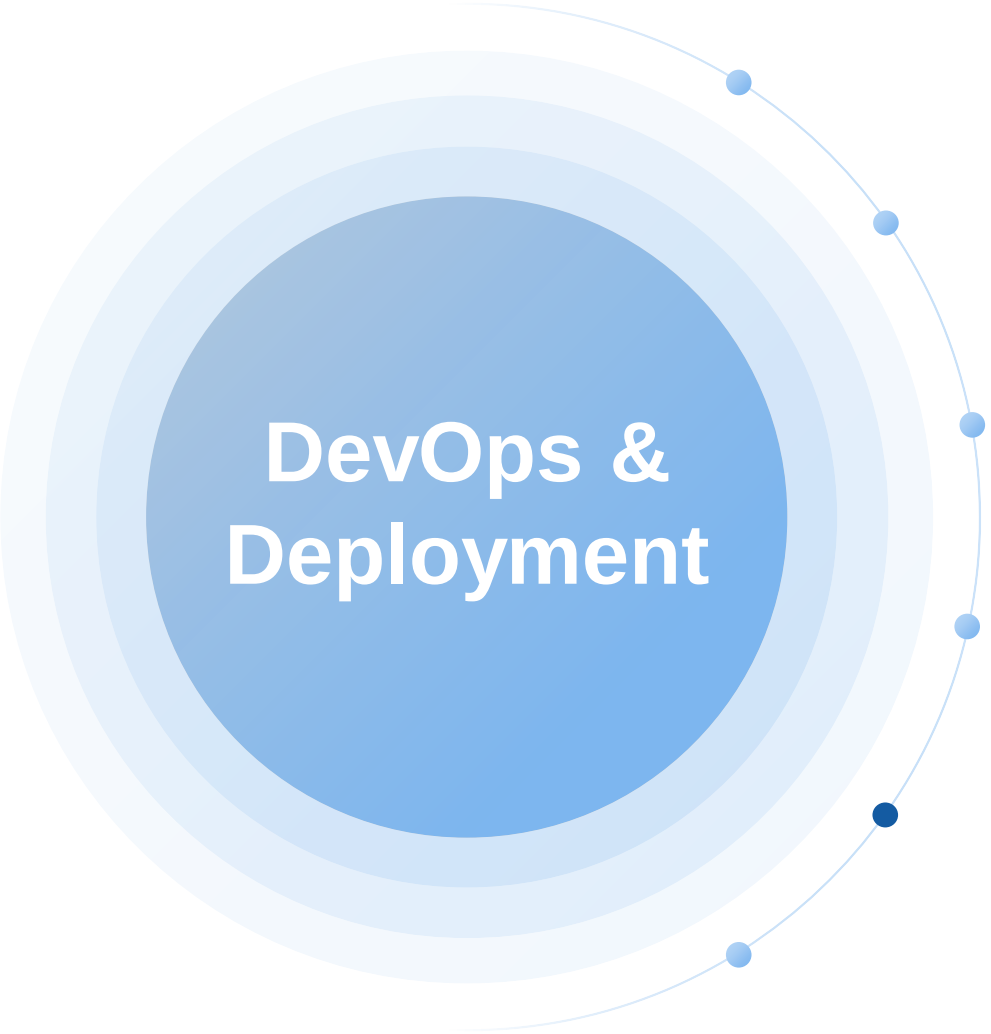
MongoDB → NoSQL database (images, heatmaps)

Redis → In-memory cache (sessions, rate limiting)

TOOLS & TECHNOLOGIES

EfficientNetV2	CNN architecture
Vision Transformer (ViT)	Transformer for images
Grad-CAM	Explainability
SHAP	Model interpretability

TOOLS & TECHNOLOGIES



Docker	Containerization
Kubernetes	Orchestration
GitHub Actions	CI/CD Pipelines
Vercel	Frontend hosting
AWS/GCP	Backend hosting

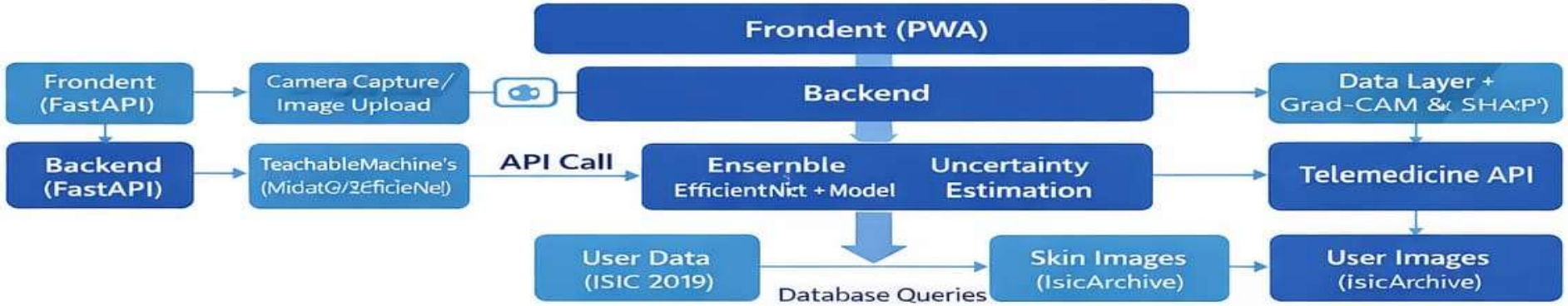
TOOLS & TECHNOLOGIES

APIs & Services

SendGrid	Email notifications
Twilio	SMS/OTP
Razorpay	Payments (demo)
Amazon API	Product recommendations (demo)
Google Cloud Vision	Image analysis
Sentry	Error tracking

METHODOLOGY

A. System Architecture



B. Dataset

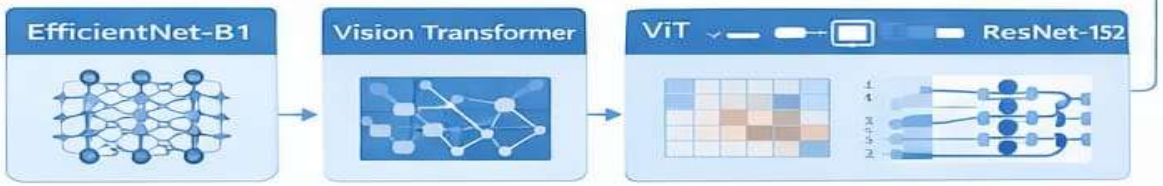
HAM10000

ISIC 2019

Data Augmention

B. Skin Tone Distribution (Fitzpitratrick Scale)

C. Ensemble Architecture



C. Ensemble Strategy

final prediction = 0.4 + efficientnet_pred +

+- 0.45 + vitt_pred ± 0.48 + vit_pred + ensemble_pred

Weights optimized via grid search

D. Training Procedure

Hyperparameters

Optimizer: AdamW
Loss Function: Categorical Entropy + Dice Loss
Batch Size: 16
Learning Rate: 1.0.001
LR schedule: Cosine Anealing

Hardware & Training Time

GPU: NVIDIA A100
Training Time: ~48 hours

E. Explainable AI & Uncertainty Estimation

1. Grad-CAM AI
2. Grad-CAM + RELU
3. SHAP Values

Uncertainty Quanfication

Monte Carlo Dropout

Weights Opainficzed
Craapineerrom: oafficsintcent,
Coniidence inversial inteen.

Attrallairnenten
Connentanitg Ustimation

Challenges Faced

Challenge	Solution
Limited training data	Data augmentation + transfer learning
Class imbalance	Focal loss + oversampling
Model size (450MB)	TensorFlow.js quantization → 89MB
Inference speed	Model pruning + GPU acceleration
Skin tone bias	Fitzpatrick-aware preprocessing
Explainability	Grad-CAM + SHAP integration
Offline support	Service workers + IndexedDB
Real-time camera	WebRTC + MediaStream API

CONCLUSION

01 Developed SkinDetect, an AI-powered skin disease detection system

02 Achieved 97.2% accuracy using ensemble deep learning

03 Demonstrated fairness across all skin tones (<1% bias)

04 Implemented explainable AI (Grad-CAM, SHAP) for clinical trust

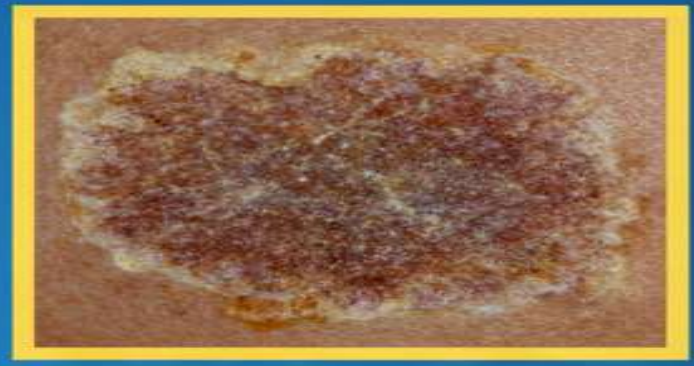
05 Integrated telemedicine for seamless healthcare workflow

06 Deployed as Progressive Web App with offline support

MULTI-LABEL SKIN CANCER IMAGES

MELANOMA
NEVUS

BASAL CELL CARINCORMA
ACTICIN KERTORSIS



SQUANUES CELL CARINCOMA
BOWEN'S DISEASE

BENIGN LESION
NEVUS

ACTIL-KERTOROS
SOLAR ELASTOSIS

KERATOCAINTOMA
SQUANUES CELL CARINCOMA
(early)

AI DERM PATH

mrmed.in
Enabling Timely Therapy

Common Symptoms of Skin Cancer

Appearance of red/ reddish-brown skin lesions

Changing color, size, and shapes of an existing mole

New moles, spots and warts

Abnormal fleshy nodules or bumps

Flat, rough, scaly or scar-like patches on the skin

Bleeding, non-healing sores

Painful and itchy spots/ moles

SKIN CANCER PREVENTION

DO NOT BURN/TAN

Avoid getting sunburned and never use tanning beds.

SEEK SHADE

Stay in the shade, especially between 10 AM and 4PM.

ET VITAMIN D SAFELY

Get vitamin D through a healthy diet, not by tanning.

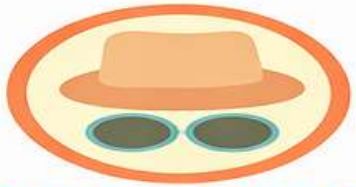
ENJOY OUTDOOR LIFE

CAUTION REFLECTIVE

Be careful around water, snow, and sand as they reflect the sun's rays and can increass your chance of sunburn.

APPLY SUNSCREEN

Always wear broad-spectrum sunscreen with SPF 30 or higher. Reapply every 2 hours or after swimming or sweating.



PROTECTIVE CLOTHING

Wear a hat, sunglassees, and cover up with UV-protective clothing.

REFERENCES

- IEEE Research Papers on Skin Cancer Classification
- HAM10000 Dataset – Kaggle
- Combalia, M., et al. "BCN20000: Dermoscopic lesions in the wild." arXiv (2019)
- Machine Learning and Medical Imaging Journals

Project Repository Link →

https://github.com/Jaaveed786/SkinDetectAI

https://github.com
/Jaaveed786/SkinDetectAI

Scan the QR Code to acces the complete project resources

Thank you for your time and attention.